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North America during the Lower Cretaceous: new palaeomagnetic constraints from intrusions in New England

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SUMMARY

Six Late Aptian and Albian intrusions in Maine, New Hampshire and Vermont, dated between 113 and 100 Ma, were sampled for palaeomagnetic analysis. The majority of samples unblock between 550 °C and 580 °C, and Curie temperatures fall in a discrete interval between 556 °C and 580 °C. This indicates that the dominant remanence carrier is a TRM and is carried by titanomagnetite, which crystallized from the magma or was produced by deuteric oxidation alteration during post-magmatic cooling. Titanomagnetite occurs as discrete C1–C3 grains, as exsolution lamellae in the silicates, and as fine reaction rims on the silicates. The palaeomagnetic results are combined with additional data from four Barremian and Early Aptian intrusions in southern Maine to produce nine reliable virtual geomagnetic poles. These possess a higher latitude than most published poles for the Lower Cretaceous. Ar^{40}/Ar^{39} ages on biotite and hornblende from these intrusions demonstrate three intrusive pulses at approximately 120 Ma, 111 Ma and 100 Ma, constraining the ages of the calculated VGPs. The corresponding palaeopoles are 72.3 °N, 198.2 °E, $A_{95} = 3.3$ °; 74.5 °N, 195.2 °E, $A_{95} = 3.8°$; and 76.6 °N, 167.5 °E, $A_{95} = 5.3°$. Based on new palaeomagnetic data reported here, significant changes to the North American Mesozoic APW path are necessary. These changes constrain the palaeoposition of the North American plate in the Lower Cretaceous.

Key words: APW (apparent polar wander), Cretaceous, magnetic mineralogy, North America, palaeomagnetism, titanomagnetite.

INTRODUCTION

The APW path for North America is well defined for parts of the Palaeozoic and Mesozoic. For some time intervals, however, due to sparse or conflicting data, the APW path is less constrained. The lack of suitable Early Cretaceous reference poles, for example, has led to speculation about a long Cretaceous standstill and subsequent rapid apparent polar wander during the Late Cretaceous.

Few palaeomagnetic studies exist on Cretaceous rocks in North America, despite the relatively long time span of 78 Myr of the Cretaceous period, and extensive outcrop. Most Cretaceous studies on the North American craton were completed in the late 1960s or 1970s (Larochelle & Black 1963; Opdyke & Wensink 1966; Hanna 1967, 1973; Larochelle 1968; Kilbourne 1969; Scharon & Hsu 1969; Shive & Fredrichs 1974; DeJournett & Schmidt 1975; Foster & Symons 1979) with no new palaeomagnetic data until 1988 (Globerman & Irving). Previous APW studies for North America by Irving & Irving (1982), Gordon, Cox & O'Hare (1984) and May & Butler (1986) all point to the critical need for additional palaeomagnetic data for the Cretaceous. The latter two compilations relied on eight studies that were summarized by Mankinen (1978). May & Butler (1986) used an average of all eight palaeopoles, but stated that 'certain of the eight poles used in this average are of questionable reliability'. Gordon *et al.* (1984) considered only four of these studies to be reliable and that the number and quality of Cretaceous poles were inadequate for a detailed analysis of the APW during the Cretaceous.

Whether the Cretaceous pole calculated by Mankinen (1978) and subsequently used by May & Butler (1986) truly represents the Cretaceous position is also subject to discussion. This time-integrated mean pole results in a standstill in the APW path for approximately 50 Myr, followed by a marked change in direction and motion of the North American plate near the end of the Cretaceous. However, given so few adequate palaeopoles for this period, the APW path remains largely unresolved. Based upon a recent study of Cretaceous and early Tertiary rocks from the North American craton, Diehl (1991), proposed that (1) little, if any, APW occurred during much of the Cretaceous, (2) a significant amount of APW (~10°–13°)

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occurred during the late Cretaceous and (3) slow APW motion prevailed once again in the Tertiary.

Several studies, including the most recent and most rigorous, have produced high-latitude poles (DeJournett & Schmidt 1975; Foster & Symons 1979; McEnroe *et al.* 1987; McEnroe & Brown 1992; Globerman & Irving 1988; Van Fossen & Kent 1992; McEnroe 1996b) for the Cretaceous. In the past, the high-latitude poles were dismissed as younger secondary magnetizations. An alternative explanation, only recently entertained, is that there was a period during the mid-Cretaceous when the pole was close to the present geographic pole. Although true polar wander (TPW) remains a possibility, there is little evidence for it at this time interval. Andrews (1985) and Livermore, Vine & Smith (1984) evaluated TPW and concluded that the late Jurassic was a time of insignificant TPW. Gordon (1987) states that TPW was so small in the last 200 Myr that it is difficult to separate it from APW. The present detailed study, with rigorous palaeomagnetic treatment and precise age control, provides new insights into these issues.

NEW ENGLAND CRETACEOUS INTRUSIONS

In order to shed light on the problems mentioned above, a palaeomagnetic investigation of 10 Cretaceous igneous intrusions from New England was carried out (Fig. 1). These complexes intrude into metamorphic and igneous basement of mid-Proterozoic to Pennsylvanian age and form part of the Cretaceous New England–Quebec intraplate magmatic province (McHone & Butler 1984). To the south-east, in the Atlantic, lie the New England Seamounts (Duncan 1984), which, when combined with the Cretaceous New England–Quebec intraplate magmatic province, comprise a region approximately 2000 km long of Cretaceous igneous activity (Foland & Allan 1992). These authors contend that magmatism was related to the effects of a stationary mantle plume that was overridden by the North American plate during the Cretaceous.

In New England little subsequent tectonic activity has occurred, nor has there been significant regional heating since the emplacement of the Cretaceous intrusions. Apatite-fission-track studies indicate this area has not been exposed to temperatures above 110 °C since the Cretaceous (Zimmerman *et al.* 1975; Miller & Duddy 1989). Vitrinite data from Mesozoic basinal sequences in Massachusetts and Connecticut further indicate temperatures less than 100 °C (Hubert, Feshbach-Meriney & Smith 1992). Therefore, thermal regional overprinting is considered unlikely, and palaeomagnetic directions are regarded as primary. Radiometric dates from biotites and hornblendes from five of the intrusions have been reported by Hubacher & Foland (1991; private communication 1991) and K–Ar dates were reported by Foland & Faul (1977).

To explore the possibility of a post-emplacement regional tilting, intrusions were chosen that lie on two lines across New England (Fig. 1). The two lines meet at the south-western tip of Maine. One line trends north for 120 km, the other west-north-west for 180 km. The intrusions sampled were ones that had been mapped in some detail (Eggleston 1918; Jenks 1934; Hussey 1962; Pierson 1970; Kanteng 1972; Johng 1975; Gilman 1978, 1979) and the results were combined with the southern Maine 125–120 Ma intrusions (McEnroe 1996b). They are composed of multiple magmatic pulses; most of these were sampled to examine possible tilting within the intrusive complexes and to ensure a time-averaged direction for the poles calculated from them.

Geology and site localities in the age range 113–111 Ma

Pleasant Mountain, Maine

Pleasant Mountain is an oval-shaped area of high ground north-west of Sebago Lake. The bedrock geological map of Maine (Osberg, Hussay & Boone 1985) indicates it as an area of feldspathoid-bearing syenite emplaced entirely within the Mississippian Sebago Granite batholith. However, the only detailed study of the petrology of Pleasant Mountain (Jenks 1934) shows that only a very small proportion of the rocks are feldspathoid-bearing, and the most abundant rock is quartz syenite (nordmarkite). The intrusion consists of a composite stock and scattered volcanic remanents. The volcanic rocks were earliest in the igneous sequence. These were followed by an intrusive phase, including analcime syenite, monzonite, diorite, augite syenite porphyry, fine-grained grey porphyritic syenite, and hybrid syenite, and then a younger phase composed of quartz syenite (nordmarkite), porphyritic hornblende syenite, fine pink porphyritic syenite, coarse anorthoclase syenite porphyry, and aplite and bostonite dykes. The late quartz syenite makes up 80–85 per cent of the entire complex. Hubacher & Foland (1991) report $^{40}Ar/^{39}Ar$ ages from biotite in a syenite of 112.1 ± 0.4 Ma.

Five palaeomagnetic sites were collected at the top of the Shawnee Peak ski area near the northern margin of the intrusion. The intrusive rocks sampled in this area are hornblende quartz syenite, (site 1) and augite syenite porphyry (sites 2, 3 and 4).

Burnt Meadow Complex, Maine

The Burnt Meadow Mountains are underlain by Cretaceous alkalic intrusive rocks enclosing large xenoliths of volcanic and/or shallow intrusive rocks. The complex, approximately 6 km long by 3 km wide (Gilman 1979), intrudes high-grade mica schists and associated migmatites and pegmatites of the Rindgemere Formation. This formation has been tentatively assigned a Lower Devonian age. Hubacher & Foland (1991) obtained $^{40}Ar/^{39}Ar$ ages from biotite from the quartz syenite of 112.6 ± 0.2 Ma.

The major rock types and their approximate relative areas of exposure include granodiorite (5 per cent), volcanic breccia and porphyry (30 per cent), syenite (20 per cent) and quartz syenite (45 per cent). Gilman (1979) interprets the sequence of emplacement for the latter three in that order, but does not comment on the granodiorite. The dominant volcanic member is a trachyte breccia. The fragmental volcanics consist of lithic fragments with a fine aphanitic matrix. The non-fragmental porphyries associated with the fragmental rocks carry phenocrysts of plagioclase, orthoclase and minor hornblende. The syenite is equigranular and is composed mainly of euhedral orthoclase (95 per cent), biotite, hastingsite, aegerine augite, fayalite, and minor titanite, titanomagnetite, ilmenite, apatite and zircon. The porphyritic quartz syenite is composed of orthoclase phenocrysts with a matrix of orthoclase (70 per cent), plagioclase (20 per cent), quartz (5–10 per cent), biotite,

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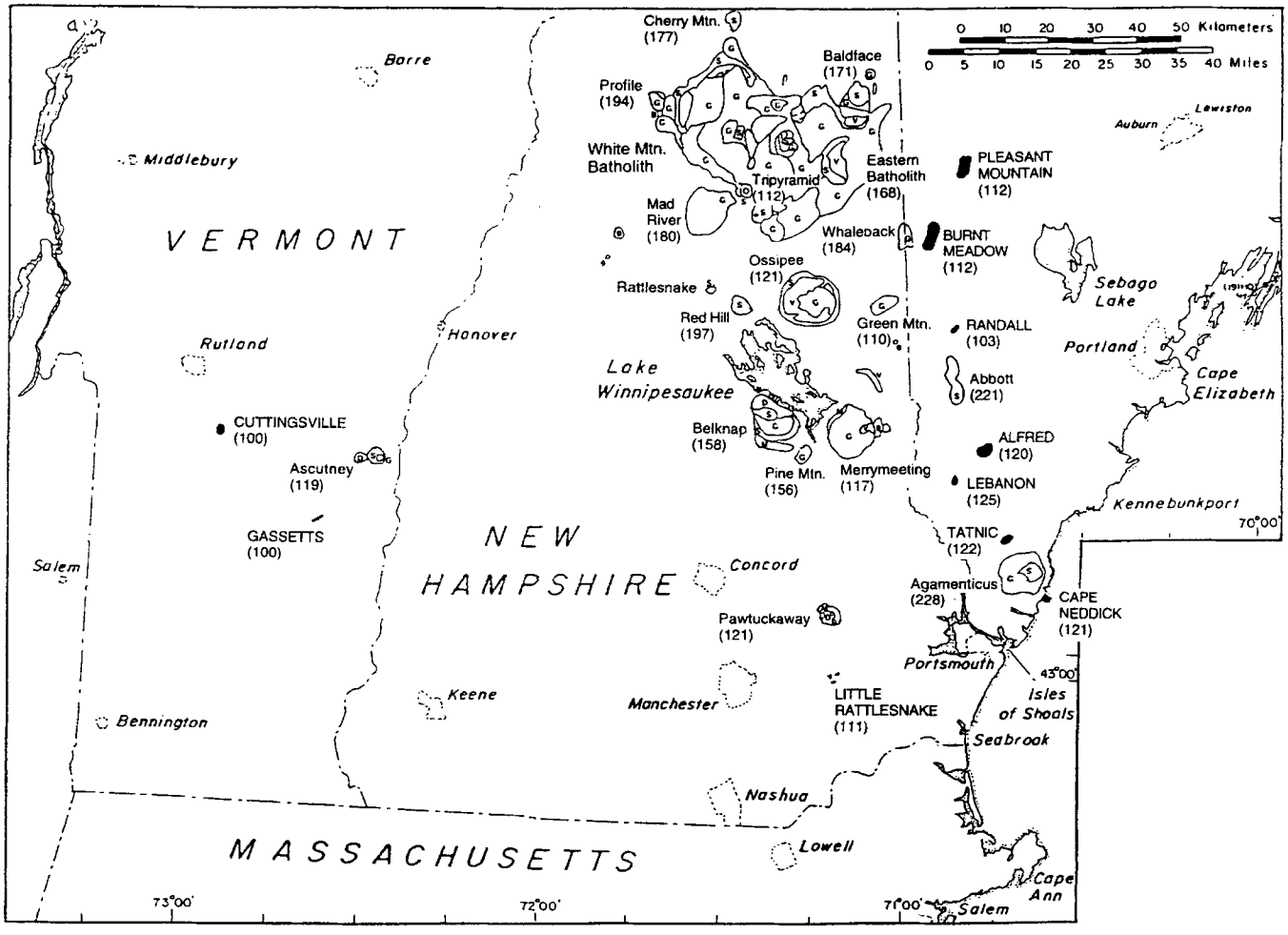


Figure 1. Map of south-western Maine and the adjacent states of New Hampshire, Vermont and Massachusetts, showing major Mesozoic intrusions (modified from McHone & Butler 1984). The Randall, Pleasant Mountain, Burnt Meadow, Little Rattlesnake and Cuttingsville intrusions, the Gassetts Lamprophyre (this study) and the southern Maine intrusions, Cape Neddick, Tatnic, Lebanon and Alfred (McEnroe 1996b), are indicated in solid black with upper-case lettering.

hornblende, and minor titanite, titanomagnetite, ilmenite, apatite and zircon.

10 palaeomagnetic sites were obtained on traverses on Stone Mountain from Dyer Pond near the centre of the complex (McEnroe 1993, Fig. 5.11). On the steep south-west face, two sites were obtained of a small mapped volcanic xenolith (sites 1 and 4), and one each of an extremely coarse porphyritic quartz syenite (2), and a normal quartz syenite (3). On a traverse up the north-west ridge, over the summit, and down the south-west ridge and face, five sites were collected from the largest volcanic xenolith (5–9) and one from normal quartz syenite (10).

Little Rattlesnake Complex, New Hampshire

The Little Rattlesnake Complex in south-eastern New Hampshire consists of three areas of volcanic to shallow intrusive rocks associated with the emplacement of gabbro (Freedman 1950; Johng 1975). These include Little Rattlesnake Hill, Hook Hill, 700 m to the north-east, and two summits of Walnut Hill, 500 and 900 m, to the south-east. The intrusion at Little Rattlesnake Hill consists of a central plug of hornblende monzonite with three mapped masses of rhyolite (Johng 1975). The hornblende monzonite is a coarse-grained rock composed of orthoclase and microperthite, plagioclase, hornblende and quartz, with accessory titanomagnetite, biotite and apatite. The rhyolite consists of microperthite, quartz, augite, biotite, magnetite and pyrite. Hook Hill is composed of andesite and andesite–rhyolite breccias consisting of a matrix of fine plagioclase, biotite, magnetite and pyrite with variable quantities of rhyolite fragments. Walnut Hill consists of two separate intrusions of gabbro and prophyritic gabbro surrounded by volcanic breccia. The gabbro consists of plagioclase, augite, orthoclase, quartz, and minor titanomagnetite, apatite and biotite. The porphyritic gabbro consists of plagioclase, orthopyroxene, orthoclase, and minor titanomagnetite, apatite and biotite. The volcanic breccia is a latite with about 50 per cent each of plagioclase and orthopyroxene, and may contain from 0 to 80 per cent of country-rock xenoliths. Foland & Faul (1977) obtained a K–Ar age on biotite in the gabbro of 111 ± 3.0 Ma.

Palaeomagnetic collections were made at 14 sites (McEnroe 1993, Fig. 5.12). Sampling on the west summit of Walnut Hill included one site in the volcanic porphyry (site 1) and one site in gabbro (site 2). Collections made on Little Rattlesnake Hill included eight sites of hornblende monzodiorite (3, 4, 5, 9, 10,

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11, 14, 15), two sites of mafic dykes (7, 12) one site of a rhyolite dyke (6) and one of rhyolitic volcanics near the southern contact (8). In addition, samples of bedded metamorphosed sandstone from the Eliot Formation were collected to the south-west of the intrusive complex for a baked-contact test.

Geology and site localities in the age range 103–100 Ma

Cuttingsville Stock, Vermont

The main Cretaceous intrusion at Cuttingsville is an oval complex of essexite, syenite, pulaskite, sodalite–nepheline syenite and quartz syenite (nordmarkite), 2.5 km long in a north-west–south-east direction and 1.8 km wide (Eggleston 1918). There are several small intrusions, including a small area of syenite and pulaskite 700 m north of the main stock, a small area of essexite 1.2 km north of the main stock, and a 950 m long mass of intrusive breccia 2.5 km north of the stock. The total area of igneous activity is about 1.8 km wide and 5.9 km long. The intrusions were mapped and described petrologically by Eggleston (1918), and more recently the petrology, petrogenesis and emplacement sequence were investigated by Pierson (1970), Kanteng (1972) and Laurent & Pierson (1972). Hubacher & Foland (1991) obtained $^{40}Ar/^{39}Ar$ ages from biotite of 100 ± 0.3 Ma.

The country rocks all belong to the middle Proterozoic Mount Holly Complex in the core of the Green Mountain anticlinorium (Doll *et al.* 1961) and consist of high amphibolite- to granulite-facies rocks that have been moderately to severely deformed and retrograded during the Taconian and Acadian orogenies. They consist mainly of hornblende, pyroxene and mica gneisses, with considerable marble and minor chloritic schists and quartzite (Eggleston 1918). There is no evidence of tectonic activity since the Devonian.

Palaeomagnetic sites were collected in a variety of rock types (McEnroe 1993, Fig. 5.15). Sites 1–4 are located on Mill River in the south-eastern part of the main intrusion. Site 1 is in mafic rock characterized as monzogabbro. Site 2 consists of essexite and diffuse grey dykes intruding the monzogabbro. Site 3 is from the dyke that cuts across the contact between monzogabbro and 'blue' syenite to the north. Site 4 is a coarse biotite syenite. Site 5 is in the breccia 2.4 km north of the main stock. The breccia is composed of quartzite, schist, gneiss and essexite porphyry in a fine-grained igneous quartzo-feldspathic matrix with abundant fine-grained dark-brown biotite. Site 6 was collected in a railroad cut 1.2 km north of the main body. The exposure is of a complex porphyry with a large xenolith of hornfelsed metamorphic rock, resembling a breccia somewhat. The porphyry has phenocrysts of plagioclase and sanidine as well as xenocrysts of quartz in a fine-grained matrix that includes hornblende and biotite. The hornfelsed xenolith is crammed with tiny equant opaque grains.

Randall Mountain, Maine

The Cretaceous intrusion at Randall Mountain (Gilman 1978) is a stock of grey–brown syenite and grey fragmental porphyritic trachyte that occurs both as inclusions and as dykes in the syenite. It is exposed in an oval area about 2.1 by 1.2 km. Exposure is poor, except on the steep slopes and cliffs of Randall Mountain near the north-western contact. The country rock consists of schists and calc-silicate rocks of the Rindgemere Formation. The syenite of the Randall Mountain stock has yielded a $^{40}Ar/^{39}Ar$ age from biotite of 103.0 ± 4 Ma (Hubacher & Foland 1991).

The syenite varies from porphyritic to medium-grained equigranular to fine grained. It is mainly composed of dusty orthoclase (95 per cent), with minor biotite, hornblende, aegerine augite, sodalite and titanomagnetite. Light-grey porphyritic trachytes are found as large inclusions in the syenite and as dykes cutting the syenite. At the north-western contact there is a wide zone of intrusive breccia, a pinkish, quartz-bearing syenite with inclusions of schist (Gilman 1978).

A total of 11 palaeomagnetic sites were sampled along two traverses, one on the south-facing cliffs near the north-western contact, and the second from the contact at the summit downwards on the south-east ridge (McEnroe 1993, Fig. 5.16). The first traverse yielded three sites of porphyry (1, 2, 4) and two of syenite (3, 5); the second yielded three sites each of porphyry (7, 8, 9) and syenite (6, 10, 11).

Gassetts Lamprophyre Dyke, Vermont

The Gassetts dyke is exposed near the southern end of the classic exposure of the Gassetts schist on the east side of Route 103, south-east of the railroad bridge at Gassetts, Vermont. The dyke strikes N44W and dips 86SW and is approximately 1.7 m thick. The dyke contains evidence of internal contacts and may be composite.

In thin section the lamprophyre is dominated by kaersutite, plagioclase and magnetite. Calcite occurs both as amygdale infills and in the matrix. The matrix also contains abundant oxychlorite after glass. A major- and trace-element analysis is reported by McEnroe (1989, Table 3.6). A K–Ar whole-rock age of 100 ± 2 Ma was determined by K. A. Foland (McEnroe 1989) and is unlikely to have substantial errors, in that the rock analysis shows 2.01 weight per cent K_2O.

PALAEOMAGNETISM

Field and laboratory procedures

Samples were collected in the field either as oriented blocks or using a portable gasoline-powered drill. Samples were oriented to $\pm 2°$ using a magnetic or sun compass. The natural remanent magnetizations (NRM) of the 272 samples were measured using a Molspin spinner magnetometer at the Paleomagnetics Laboratory, University of Massachusetts, Amherst, or a JR-5 spinner magnetometer at the Paleomagnetics Laboratory of the Norwegian Geological Survey (NGU), Trondheim. At least two specimens from each site were selected for detailed alternating-field (AF) and/or thermal-demagnetization treatment. For AF treatment, specimens were progressively demagnetized at increments of 5 or 10 mT up to a peak induction of 100 mT using a three-axis tumbling demagnetizer (NGU) or a single-axis non-tumbling Schoenstedt GSD-1 AF Geophysical specimen demagnetizer (UMASS). The AF demagnetization method proved effective in isolating the primary remanence. Remaining samples were AF-demagnetized in three or more increments.

For thermal demagnetization carried out at NGU, specimens were heated in air in steps of between 50° and 100 °C to temperatures of 585 °C or 630 °C using a non-inductively wound furnace with an accuracy of ± 5 °C, cooled in a field-free

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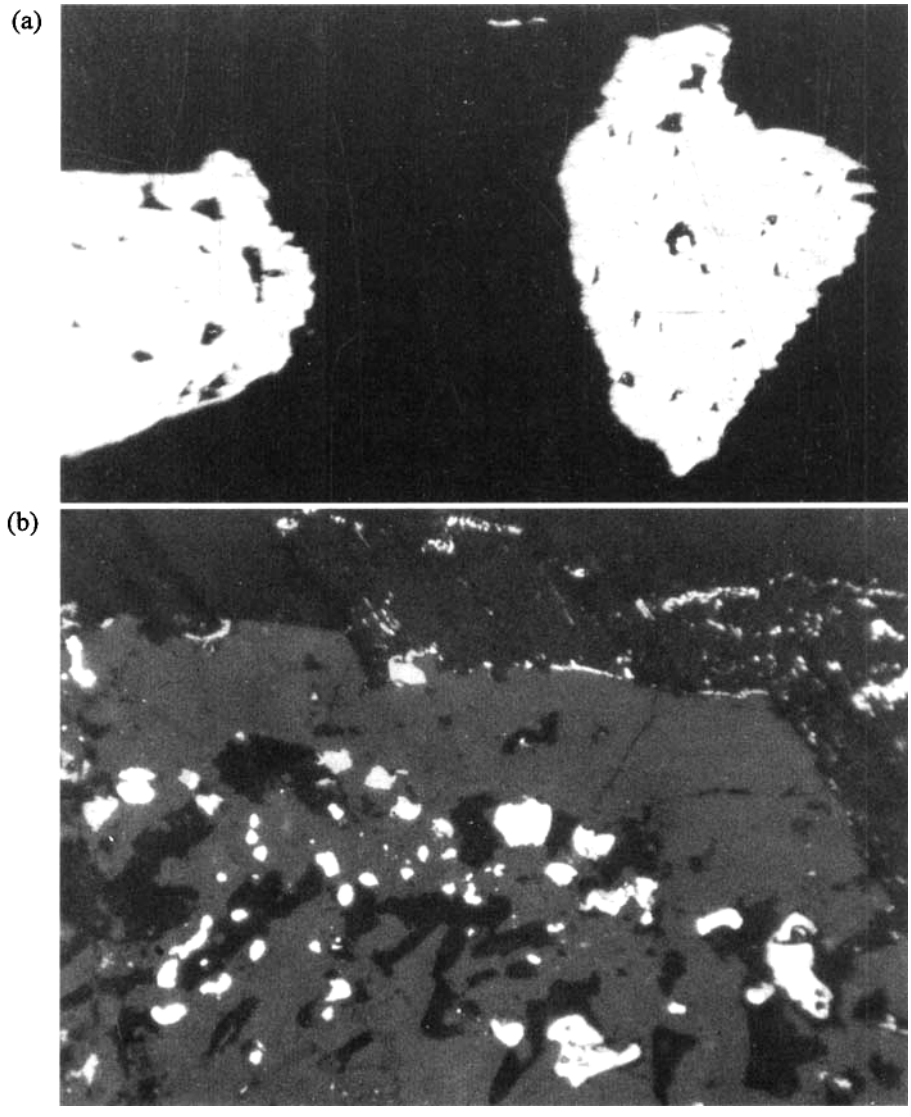


Figure 3. (a) Photomicrograph of C3 titanomagnetites from the essexite phase of the Cuttingsville intrusion (sample Cut-2F). The sandwich-type ilmenite lamellae are developed along {111} planes of the titanomagnetite. Inclusions of pyrite (bright white) and silicates (black) are present. (b) Photomicrograph of silicates from the essexite phase of the Cuttingsville intrusion (sample Cut-2F). Abundant inclusions of magnetite and ilmenite in pyroxene. Magnetite forms as a rim between the pyroxene and Ti-rich biotite, and as a rim on the biotite. Magnetite and ilmenite are developed from the biotite and are concentrated along cleavage planes. Field of view 300 μ. Photographed with a blue filter.

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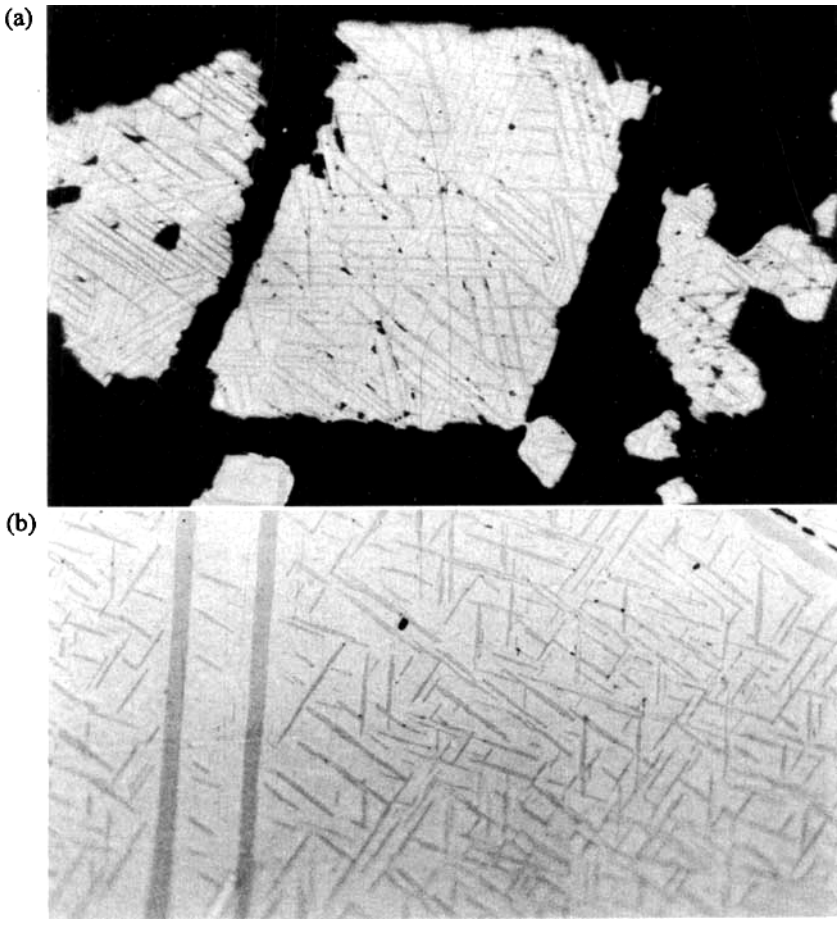


Figure 5. (a) Photomicrograph of titanomagnetites from the gabbro in the Little Rattlesnake intrusion (sample LRS-1). Titanomagnetite grains have very fine {111} trellis oxidation exsolution of ilmenite (grey) lamellae. Field of view 300 μ. Photographed with a blue filter. (b) Photomicrograph from the volcanic porphyry, Randall Mountain (sample RM2-1). Titanomagnetite grain with very fine {111} trellis oxidation exsolution of ilmenite (grey) lamellae and with fine granular exsolution of pleonaste (black). Field of view 50 μ. Photographed with a blue filter.

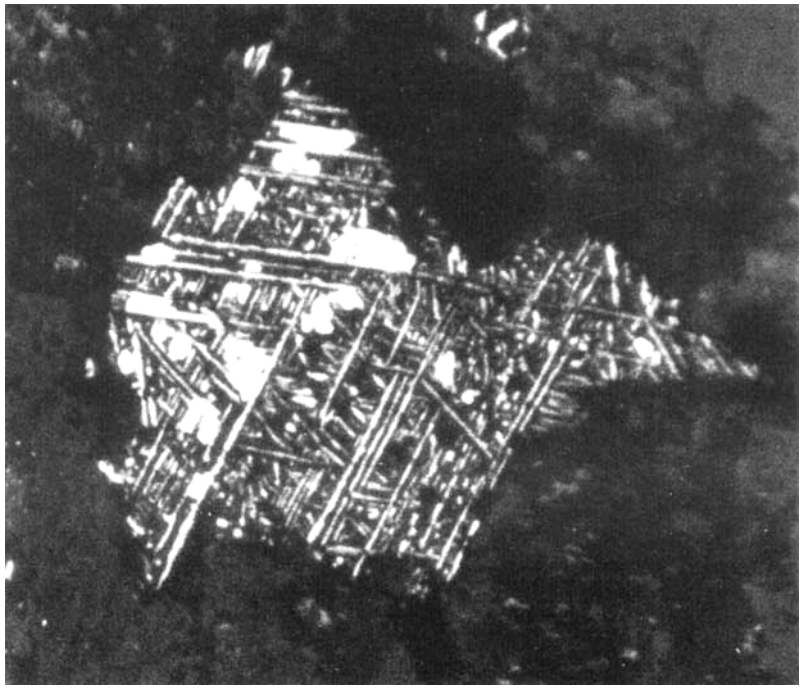


Figure 9. Photomicrograph from the hydrothermally altered zone in the Little Rattlesnake Complex (sample LRS). The magnetite portion of the titanomagnetite has been dissolved away and in part replaced by pyrite. The ilmenite lamellae are unaltered and show the relic {111} planes of oxidation–exsolution from the original titanomagnetite grain. Discrete grains of pyrite in the matrix are also shown.

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chamber at NGU and measured on a magnetometer in an area with an ambient field of < 5 nT.

Data were plotted on standard orthogonal vector-end-point diagrams (Zijderveld 1967). The primary remanent magnetization direction was determined by standard palaeomagnetic techniques, or, when necessary, Spectrum (Smethurst 1987). Site mean directions and confidence limits were calculated using Fisher (1953) statistics. The virtual geomagnetic pole (VGP) for each intrusion was calculated by Fisher statistics on site mean VGPs. Mean poles for given age intervals were calculated by averaging the virtual geomagnetic poles for all intrusions in a given age range.

Magnetic domain structures were investigated by measuring hysteresis properties and by the use of demagnetization tests of the NRM and IRM. To induce saturation, samples were subjected to fields of up to 1.25 Tesla (T) in 12 increments and then demagnetized. Hysteresis properties and Curie temperatures were measured on a Princeton Applied Research Vibrating Sample Magnetometer at the Institute for Rock Magnetism, University of Minnesota. Hysteresis curves were measured at room temperature in a maximum field of 0.8 T.

Optical observations were combined with measured rock magnetic properties and demagnetization behaviours of samples to assess the reliability of the signal and to assess the timing of acquisition of magnetization. Observations of the opaque mineral size, morphology, texture, degree of oxidation and exsolution features were made using oil immersion and reflected light.

Remanence behaviour

113–111 Ma intrusions

Palaeomagnetic data from 15 sites (161 samples) from the Burnt Meadow and Pleasant Mountain intrusions in Maine, and the Little Rattlesnake intrusions in New Hampshire are listed in Table 1. The NRM intensities of these samples range from 0.1–2.4 A m^{-1}. During AF demagnetization, samples

Table 1. Summary of mean palaeomagnetic data from Aptian New England intrusions 113–111 Ma.

Burnt Meadow Mountains, Maine (70.92°N, 43.85°E)

Site	N/N_O	D	I	a95	k	Plat °N	Plong °E
BM 1	7/7	357	60	4.1	213	86	147
BM 2	7/7	324	66	5.0	145	65	222
BM 3	5/6	350	60	4.0	371	82	181
BM 4	4/4	339	74	5.6	269	69	259
BM 5	7/7	334	63	4.0	228	71	210
BM 6	6/6	337	65	5.4	157	74	218
BM 7	3/3	338	62	5.3	544	74	205
BM 8	4/4	352	57	2.4	1454	81	156
BM 9	4/5	321	60	6.9	177	61	207
BM 10	5/5	336	70	5.3	209	72	240
Site Mean	10/10	340	63	5.6	98	75.7	208.5

Pleasant Mountain, Maine (70.82°N, 44.03E)

SITE	N/N_O	D	I	a95	k	Plat °N	Plong °E
PM 1	5/5	339	58	6.8	127	73	188
PM 2	7/7	328	58	3.2	349	66	198
PM 3	8/8	339	59	8.3	46	74	191
PM 4	6/6	360	58	4.3	247	85	109
PM 5	7/7	355	60	10.3	36	85	161
Site Mean	5/5	344	59	6.5	141	77.4	185.0

Little Rattlesnake Intrusions, New Hampshire (71.08°N, 42.98°E)

SITE	N/N_O	D	I	a95	K	Plat °N	Plong °E
LRS 1	9/9	347	47	18.0	72	73	148
LRS 2	6/6	322	52	5.1	206	59	193
LRS 3	5/5	342	58	4.1	342	76	187
LRS 4	6/6	336	52	5.2	168	69	178
LRS 5	6/6	354	61	6.7	102	86	189
LRS 6	5/5	338	45	6.5	138	66	164
LRS 7	4/4	325	62	3.6	654	65	212
LRS 8*	4/4	59	28	27.0	69	32	29
LRS 9*	6/6	260	51	22.2	10	14	229
LRS 10	6/6	346	58	5.7	139	79	182
LRS 11	8/8	327	52	6.9	65	62	188
LRS 12	7/7	340	57	2.6	521	74	186
LRS 13	4/4	336	64	6.1	229	73	216
LRS 14	6/6	338	47	8.1	69	67	166
Site Mean	14/14	340	57	11.0	14	73.9	186.2
Selected Mean*	12/14	338	55	4.4	96	71.5	182.6

N/N_O: ratio of samples used in determining site mean directions to the number of samples demagnetized; D, I: mean declination and mean inclination; a_{95} and k: Fisher (1953) statistical parameters; Plat, Plong: latitude (°N) and longitude (°E) of magnetic pole; mean: mean at site level; mean*: selected mean excludes sites with unstable magnetization.

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showed the presence of a 'soft' component, which was removed by 20 mT (Fig. 2). At higher fields, a component with a linear decay heading towards or reaching the origin in a north-west to north-east and down (positive) direction is present (Fig. 2). The coercivity of the specimens varied from highest values in the volcanics with a medium destructive field (MDF) of 45 mT to lowest in the syenites (MDF = 10 mT). Although the average MDF from these samples is lower than in the 100 Ma group of intrusions studied, a primary magnetization was successfully isolated. Even though there are various coercivities in the

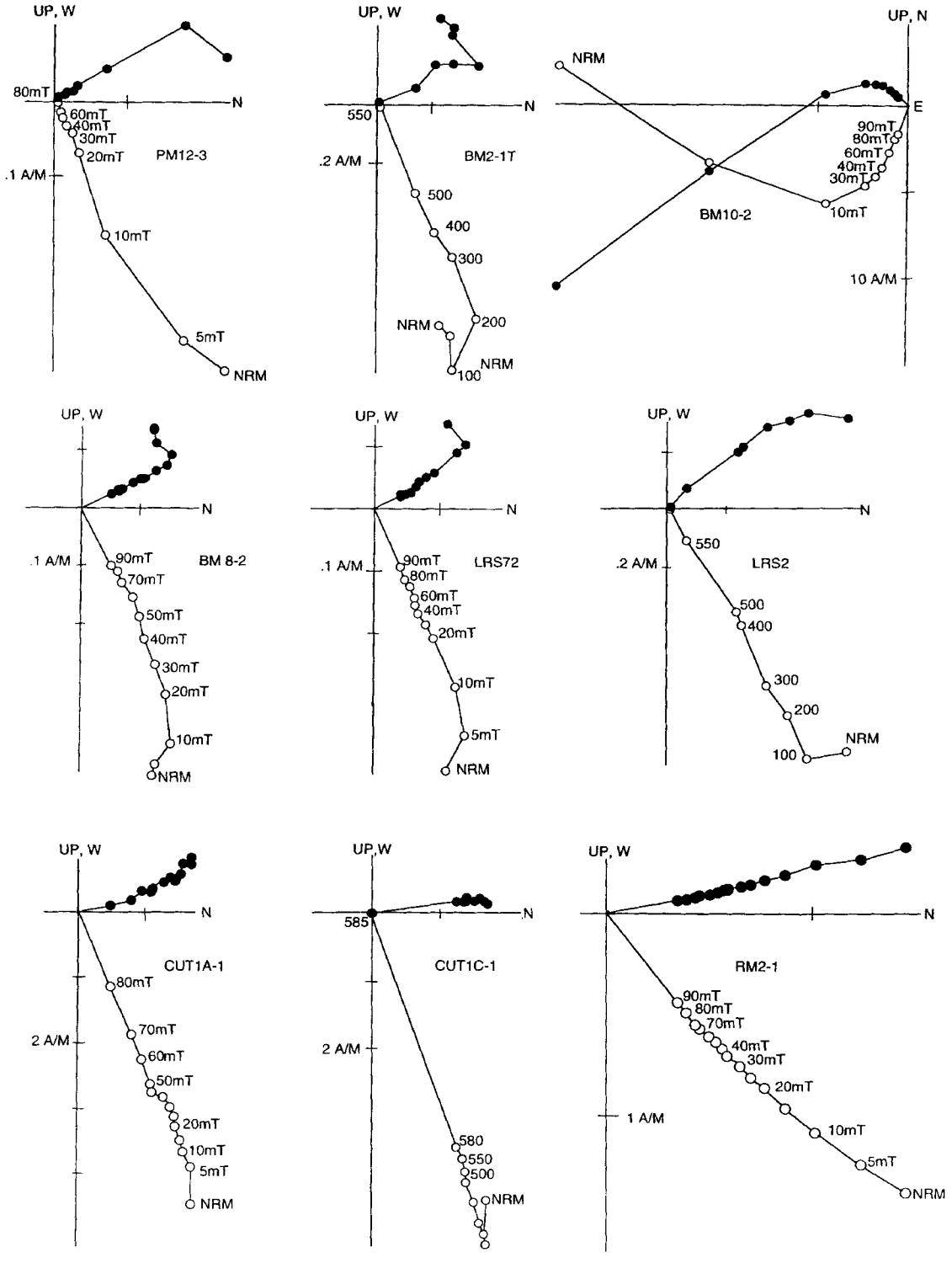


Figure 2. Vector-end-point diagrams for samples from the Pleasant Mountain (PM), Burnt Meadow (BM), Little Rattlesnake (LR), Cuttingsville (CUT) and Randall Mountain (RM) intrusions. W = west, E = east, S = south and N = north. Closed circles: horizontal component; Open circles: vertical component; numbers adjacent to circles: alternating fields in mT or temperature (°C) for thermal demagnetization. Origin included in vector.

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samples from the different members of the intrusive complexes, there was little change in direction above 20mT. In a few samples from the Little Rattlesnake complex, a primary remanent magnetization could not be determined. Petrographic observations showed that there was a local zone of hydrothermal alteration.

103–100 Ma intrusions

A total of 111 samples from the 18 sites from the Cuttingsville, Randall Mountain and Gassetts intrusions were analysed (Table 2). The NRM directions of these samples were north to north-west and down (positive). NRM intensities ranged from 0.5 A m^{-1} to 4.7 A m^{-1}. Only a minor overprint is present in these samples. Under AF demagnetization, after the removal of a secondary component at low fields, there is a well-defined component with a univectorial decay heading towards or reaching the origin in a north-westerly and down (positive) direction (Fig. 2). Sites from the essexite portion of the Cuttingsville intrusions in Vermont and the volcanics from the Randall Mountain intrusions in Maine show a hard coercive component with MDFs of 30–50 mT. The volcanics at Randall Mountain have >40 per cent of the magnetization remaining above 50 mT and >30 per cent after peak inductions of 90 mT. The characteristic remanent directions obtained from the samples that carry this hard magnetization are similar to those in other samples of lower coercivity (MDF 10–30 mT).

Magnetic mineralogy and stability

Given the possibility of a high-latitude North American pole in the Cretaceous (DeJournett & Schmidt 1975; McEnroe *et al.* 1987; Globerman & Irving 1988; McEnroe & Brown 1992; Van Fossen & Kent 1992; McEnroe 1996b), it is necessary to examine mechanisms that could modify the palaeomagnetic results or invalidate the data from which a high-latitude pole is produced. Contamination from the present-day or recent geomagnetic field could lead to a misinterpretation of palaeomagnetic data. To avoid such problems it is necessary to examine mechanisms that could produce a high-stability recent overprint. In this study 267 samples were subjected to detailed AF or thermal demagnetization. Could these rocks have been overprinted by relaxation of magnetic domains, by a thermal process, or by a chemical process, in a manner that is difficult to detect? To address this question, polished cores and thin sections were examined to identify the magnetic carriers and to note any alteration that could result in a change in the remanent magnetization. Optical observations were combined with measured rock-magnetic properties and demagnetization behaviours to assess the reliability of the signal and the timing of magnetization acquisition.

Table 2. Summary of palaeomagnetic data from the 103–100 Ma New England intrusions.

Cuttingsville Intrusions, Vermont (43.50°N, 72.87°E)

SITE	N/N_O	D	I	a_{95}	k	Plat °N	Plong °E
CUT 1	8/8	342	62	4.6	169	77	202
CUT 2	8/8	348	61	5.6	111	81	192
CUT 3	8/8	341	60	5.1	120	76	193
CUT 4*	3/3	332	59	24.0	31	69	197
CUT 5	10/10	342	55	3.5	219	74	173
CUT 6	6/6	306	61	4.2	256	51	214
Site Mean	6/6	336	60	6.5	107	72.4	200.7
Selected Mean*	5/6	336	61	8.3	87	72.1	197.0

Randall Mountain, Maine (43.67°N, 70.83°E)

SITE	N/N_O	D	I	a_{95}	k	Plat °N	Plong °E
RM 1	5/5	347	55	7.0	122	77	165
RM 2	8/8	348	53	5.8	93	76	156
RM 3	5/5	360	53	4.0	372	80	109
RM 4	6/6	349	46	5.1	174	71	141
RM 5	8/8	352	46	3.3	283	73	134
RM 6	6/6	359	51	2.6	688	78	113
RM 7	4/4	359	47	8.2	128	75	113
RM 8	3/4	330	53	9.5	168	65	186
RM 9	6/6	352	54	8.9	58	79	146
RM 10*	6/6	45	45	6.8	98	50	29
RM 11	8/8	2	52	2.6	465	79	101
Site Mean	11/11	357	52	7.0	241	78.7	122.2
Selected Mean*	10/11	352	51	4.0	145	76.5	140.0

Gassetts, Vermont (43.3°N, 72.59E)

Site	N/No	D	I	a_{95}	k	Plat °N	Plong °E
Gassett 1	7/7	341	56	10.9	32	74.0	178.9

N/N_O: ratio of samples used in determining site mean directions to the number of samples demagnetized; D, I: mean declination and mean inclination; a_{95} and k: Fisher (1953) statistical parameters; Plat, Plong: latitude (°N) and longitude (°E) of magnetic pole; mean: mean at site level; mean*: selected mean excludes sites with unstable magnetization.

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Petrographic and rock magnetic observations related to magnetic stability

When titanomagnetite grains are subjected to oxidation–exsolution of ilmenite, the remaining magnetite is more Ti-poor and hence has a higher Curie temperature than the original titanomagnetite. The oxidation–exsolution process may also produce a smaller effective magnetic grain size because the ilmenite lamellae parallel to the {111} planes of magnetite isolate the areas of Ti-poor titanomagnetite. These new compositions and textures produced by oxidation–exsolution may have different rock magnetic properties. The oxidation states of the observed titanomagnetites fall into classes C1 to C3 of Haggerty (1976). Ilmenite intergrowths in the titanomagnetites are classified into three types: sandwich, trellis and composite (Buddington & Lindsley 1964). The morphologies of the titanomagnetites range from skeletal to cruciform and herring bone to euhedral.

In Fig. 3(a), a photomicrograph of an essexite sample from the Cuttingsville intrusion (site 2), most of the titanomagnetite grains observed are significantly larger than the 15 μm PSD threshold and are hence MD-sized structures. This is surprising because this sample displays a hard coercivity under AF demagnetization (Fig. 4a) with a MDF = 60 mT, implying PSD structures for many of the titanomagnetite grains.

C2 titanomagnetites with {111} oxidation–exsolution lamellae of ilmenite are the typical magnetic grains in the Little Rattlesnake intrusions (Fig. 5a) and in the volcanics of the Randall Mountain intrusions (Fig. 5b). Grain sizes of titanomagnetite range from 0.5 mm down to the optical resolution of the microscope (about 0.4 μm). Even the fine-grained titanomagnetites contain abundant trellis {111} oxidation–exsolution lamellae of ilmenite. Optically, the majority of these grains would be classified on the basis of size as MD, and therefore unstable carriers of remanent magnetization over the geological time-scale of interest. This contradicts the bulk measurements showing that the field needed to remove the NRM is usually a medium to hard coercive force ranging from 30 mT to greater than 90 mT. In Fig. 4(a), samples from Randall Mountain (RM 1–3) and Little Rattlesnake (LRS 1–1) indicate MD behaviour early on, as shown by a rapid decay of the NRM, and then display a hard coercive component, as shown by a steady decay curve. Were some of the 'MD-sized' grains in fact the 'effective' remanent magnetic carriers? Larson *et al.* (1969) first proposed that the subdivision of titanomagnetite grains by oxidation could generate a SD-type behaviour. In earlier work, Shive & Butler (1969) dismissed the subdivision of grains in the titanomagnetite series by oxidation–exsolution as a possible mechanism for magnetic stability. However, in the hemo-ilmenite series, they demonstrated that the subdivision of the grains by exsolution played an important role in magnetic stability. Özdemir, Dunlop & Moskowitz (1993) have shown that oxidation processes may increase internal stress in the titanomagnetite grain, which in turn would impart a higher coercivity to the oxidized titanomagnetite.

One way to address the question of magnetic stability in the titanomagnetite series is by evaluating the percentage of intensity remaining after AF demagnetization to 30 mT, a reasonable field for the demagnetization of titanomagnetite grains down to ~6 μm size and stronger than the threshold for the ~15 μm 'true MD' grains, and then relate this to the original NRM intensity and the percentage of titanomagnetite that is

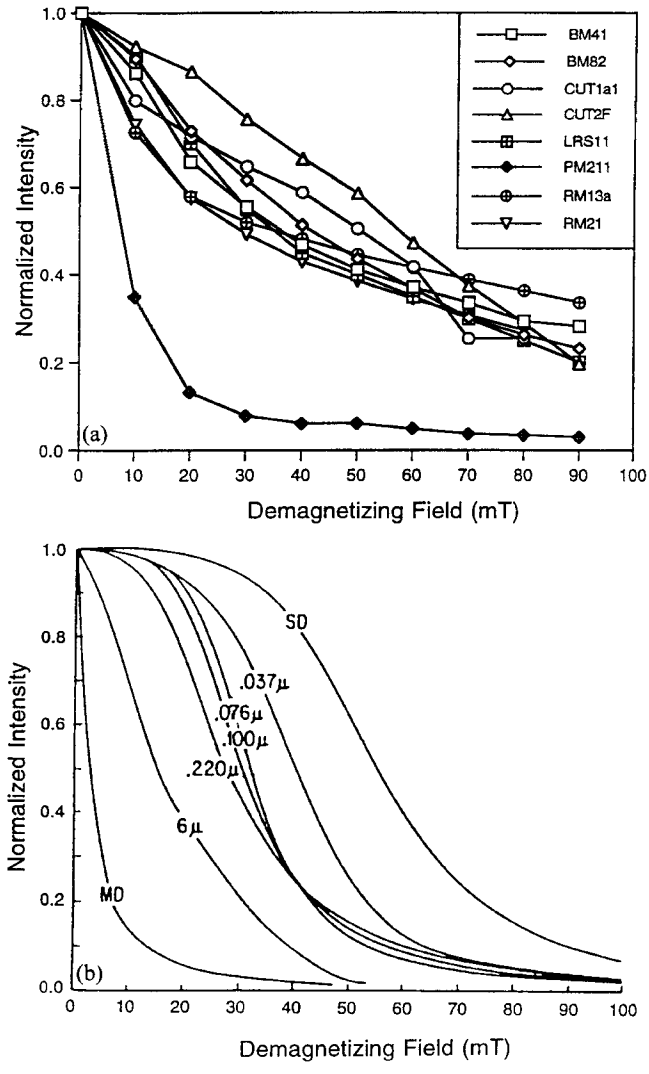


Figure 4. (a) Normalized NRM demagnetization curves for representative AF samples from Cretaceous (113–100 Ma) intrusions. (b) AF demagnetization curves of TRM in magnetites, modified from Dunlop (1973) and Parry (1965).

in the thin section. Examples from the Randall Mountain, Burnt Meadow Complex and Little Rattlesnake intrusions all show a significant component of the remanent magnetization remaining after being subjected to 30 mT, as shown in the stability diagrams (Fig. 6). Assuming equal saturation of the magnetite grains, if the stable remanence is carried only by grains with a diameter less than 15 μm then there must be 70 per cent of these grains in the samples. Since the amount of visible magnetite in thin sections is about 4 per cent, an additional 9 per cent of optically undetected fine-grained magnetite must exist, giving a total of 13 per cent magnetite. Given the composition of these rocks, a total of 13 per cent magnetite is unreasonably large from a petrological stand point.

An alternative explanation may be that some of the true MD-sized grains act as PSD grains. Though the titanomagnetite grains are large, the magnetite areas between the trellis lamellae are quite small (<2 μm), and it may be that these very fine trellis oxidation–exsolution features (Fig. 5b) exist on an even finer scale, creating stable PSD or SD-sized grains

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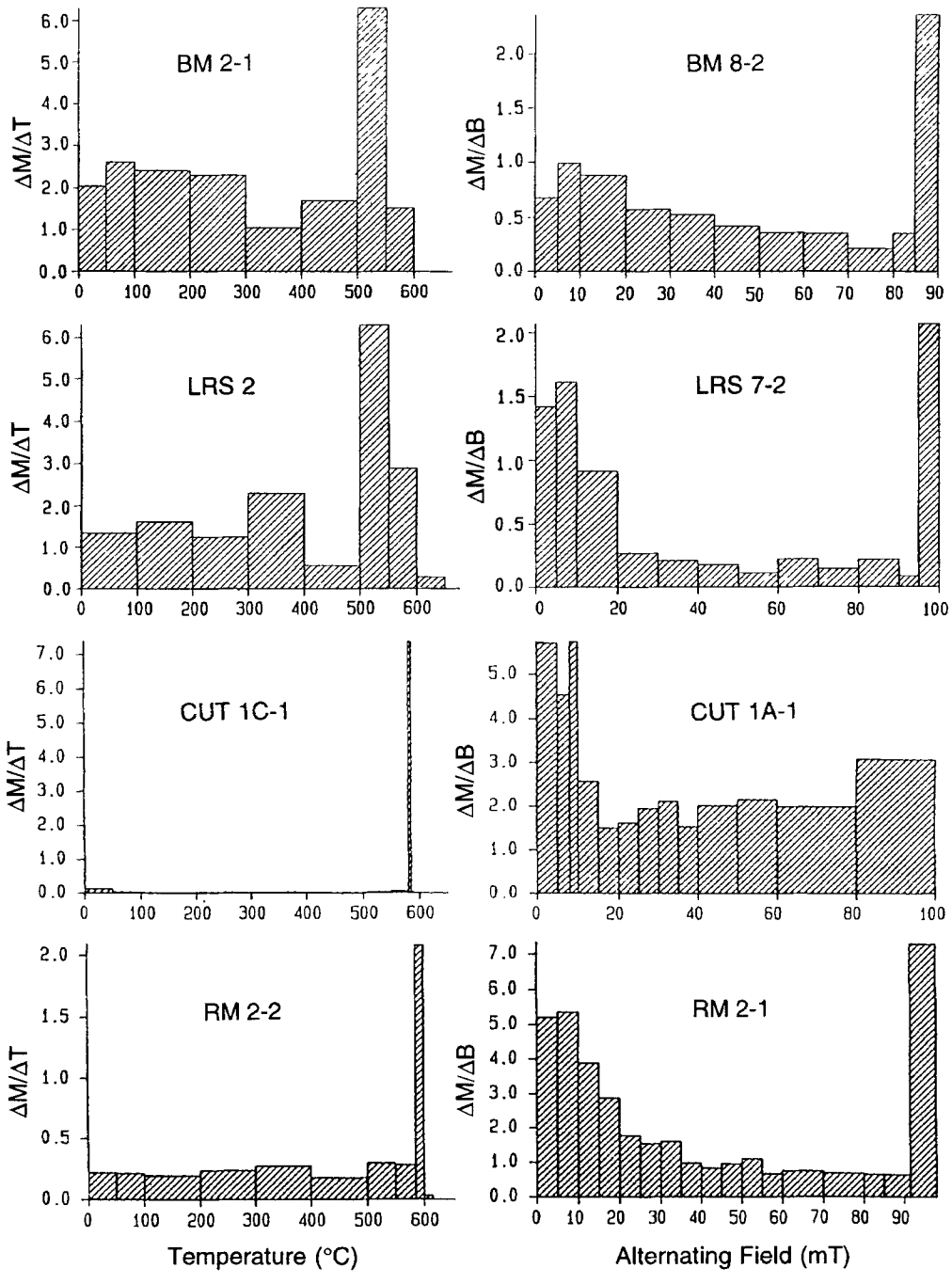


Figure 6. Magnetic stability diagrams for representative samples from the Burnt Meadow, Little Rattlesnake, Cuttingsville and Randall intrusions. Right: AF stability. ΔM/ΔB: change in magnetic moment/change in applied field in mA mT^{-1}. Left: Thermal stability. ΔM/ΔT: change in magnetic moment/change in temperature in mA $°C^{-1}$.

within the larger MD grains. If the stable remanence is carried in the magnetite-rich areas between the trellis lamellae, then shape anisotropy may also be playing a role and enhancing the coercivity of the remanence.

Magnetic relaxation

Larger grains in coarse-grained igneous rocks have not been considered as likely carriers for stable remanence because of the mobility of domain walls. However, Moon & Merrill (1986) proposed that if the domain walls in large grains are pinned with dislocations controlled by the internal field, there could be a resultant remanence. Xu & Merrill (1992) further proposed that the temperature stability of bulk coercivity of all MD grains is predominantly controlled by the magneto-elastic energy associated with dislocations. Conversely, Halgedahl & Fuller (1983) suggested that some MD-sized grains may fail to nucleate any domain walls and hence remain in a single-domain state that could consequently play an important role in the stability of the TRM. In general, 'true MD-sized' grains would have a magnetic relaxation time far less than the 120 Myr of geological interest here.

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Pseudo-single-domain grains have a proposed diameter range of ~0.1–~15 μm (Dunlop 1990), up to a threshold of ~40 μm (Day 1977), although Moskowitz & Banerjee (1979) would limit this to 8 μm. PSD grains can show a significant coercivity and stability of NRM and may play an important role in the retention of the magnetic signature in igneous rocks.

In Fig. 4 the NRM demagnetization curves are compared to the TRM curves of Dunlop (1974, 1981) and Parry (1965). These curves show the relationship between grain diameter and the shape of the TRM curve, here substituted by the NRM demagnetization curve (a weak-field TRM). Most of the demagnetization curves display predominantly PSD- and SD-type behaviour. The remaining samples show a mixed behaviour that indicates predominantly PSD grains with some MD grains. Samples from Randall Mountain (RM 1–3), and Little Rattlesnake (LRS 1-1) indicate some MD behaviour early on, as shown by a rapid decay of the NRM, then display a hard coercivity as shown by a steady decay curve. All samples have unblocking temperatures in the magnetite spectrum, indicating magnetite as the predominant magnetic carrier.

IRM acquisition curves (Fig. 7a) show the acquisition of magnetic saturation in most samples by applied fields of 0.2 T. This supports the AF and thermal-demagnetization data that indicate titanomagnetite is the predominant magnetic carrier. The two Burnt Meadow samples (BM2–2A and BM1–2) do not saturate so easily at low fields. The Burnt Meadow samples contain feldspars with opaque inclusions that could be responsible for the harder saturation. The Randall Mountain volcanics sample RM1–2 displays a gentler gradient below 0.5 T and saturation is not achieved until 0.6 T. In this sample opaque grains are present as exsolution lamellae in the silicates. Given the crystallographic controls that produce elongate or platey crystals, these exsolved grains may be harder to saturate than the discrete magnetites. Although many of these grains are larger than the PSD threshold, they could be acting as PSD grains because of the highly elongated shape (Dunlop 1990).

Demagnetization of IRM-saturated samples can distinguish between SD, PSD and MD grains (Dunlop 1972, 1983; Volk 1977). For rocks of predominantly single-domain particles, demagnetization of the saturation IRM is easier than the demagnetization of the natural remanent magnetization. In contrast, MD particles are more resistant to the demagnetization of the IRM than the NRM. In comparing the IRM-normalized demagnetization curves in Fig. 7(b) to the normalized NRM curves in Fig. 4(a) it is surprising that all samples show a stronger resistance to demagnetization of the NRM than of the IRM, thus showing that a substantial component of the remanent magnetization is behaving as if contained in SD-like or PSD-like particles.

To examine this possibility, the experimental work by Dunlop (1983) on domain structures was used. Dunlop chose the parameter $\Delta H_{1/2}/H_{1/2}$, where ΔH is the difference between the median destructive fields (MDFs) of the weak field (NRM) to strong field (IRM). This ratio is normalized to the median destructive field of the NRM (weak field). Dunlop observed that multidomain grains have negative values of $\Delta H_{1/2}/H_{1/2}$, whereas SD and PSD grains have positive values. The SD and PSD have values of +0.05 to +0.24, and mixed populations show a slight overlap, though mostly a range of higher positive values between +0.3 and +0.6. The values calculated from these samples indicate that all three types of grains are present,

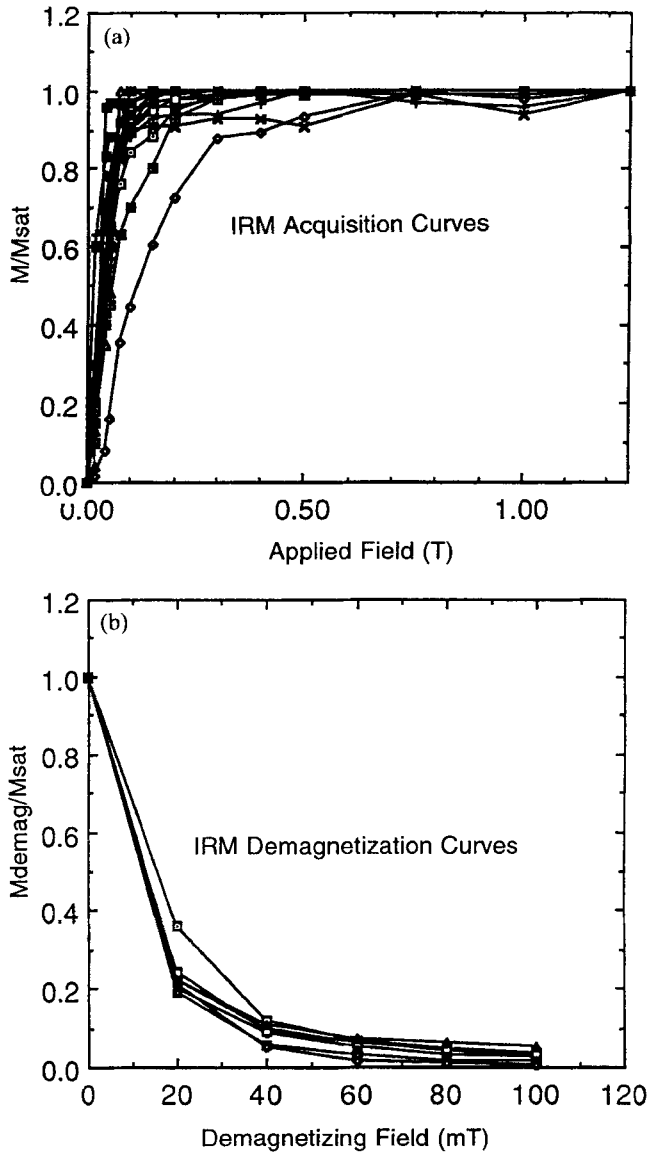


Figure 7. (a) Isothermal remanent magnetization acquisition curves. M/Msat: magnetization applied normalized to saturation magnetization. Field is in T. (b) Demagnetization of isothermal remanent magnetization. Mdemag/Msat: demagnetization normalized to saturation magnetization. Field is in mT.

SD, PSD and MD. In most samples measured, the results indicate predominantly mixed populations, with values of +0.17 to 0.75 (McEnroe 1993). This range is in good agreement with hysteresis properties measured on samples from the intrusions, which, in a plot of *J*rs/*J*s versus *H*rc/*H*c fall into all three fields, SD, PSD and MD. *J*rs/*J*s ranged from 0.2 to 0.6 and *H*rc/*H*c from 1.0 to 7.0, but plotted predominantly in the lower corner of the PSD region. Given that these are bulk-rock measurements, it is the coercivity of the sample that is measured and will reflect the mixed population of grains.

From the above observations and discussions of petrography and rock-magnetic properties, one can conclude that, although these are 'coarse-grained' igneous rocks, most samples have retained a reliable magnetic memory. Furthermore, the morphology and/or oxidation–exsolution textures of the grains

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may be playing as important a role as grain size in the coercivity of remanence.

Thermal remanence

Unblocking temperatures were typically between 500 °C and 580 °C, indicating that titanomagnetite is the primary remanence carrier, and reflecting compositions with 0 to 20 per cent ülvospinel in solid solution. The stability diagrams (Fig. 6) show variations in thermal unblocking temperatures. Some samples show one discrete unblocking temperature between 580 °C and 585 °C (CUT 1C-1, Fig. 6). The photomicrograph of this coarse-grained diorite (Fig. 3a) shows titanomagnetite grains that are in the C3 oxidation stage with sandwich-type lamellae. The unblocking temperature indicates that virtually all of the titanium has diffused out of the original titanomagnetite, leaving a pure magnetite grain with ilmenite oxidation–exsolution lamellae. This is also in agreement with the sandwich-lamellae texture. A single blocking temperature also implies that all the magnetite grains are of approximately the same composition. In contrast, under AF demagnetization sample Cut 1A-1 yields a gradient of coercivities (Fig. 6). Titanomagnetite and ilmenite grains are present in different morphologies and as inclusions in the silicates (Fig. 3b), which could account for variation in coercivity but, as indicated by thermal demagnetization, a uniformity in composition.

In a very few samples, unblocking temperatures were present above the magnetite spectrum, up to 630 °C. The direction obtained from the higher-temperature part of the decay path was similar to that obtained in the lower-temperature part (below 580 °C). Based on the unblocking temperature, this component is most likely carried by haematite, which may be present as exsolution lamellae in the feldspars and/or pyroxenes.

Curie temperatures

Curie temperatures were measured on a Vibrating Sample Magnetometer using rock chips or powders, with results listed in Table 3. Except for the sample from the Little Rattlesnake intrusion, discrete Curie temperatures in the narrow interval range from 556 °C to 572 °C were measured, indicating Ti-poor titanomagnetite. These samples produced reversible heating and cooling curves (Fig. 8). The Little Rattlesnake intrusion sample showed two inflections: a low-temperature one at 350 °C, and a high-temperature one at 580 °C. The low-temperature fraction may be carried by an intergrowth of pyrrhotite and titanomagnetite, and the high-temperature fraction by a near-end-member C2 titanomagnetite that contains fine abundant exsolution of {111} ilmenite lamellae. The range in Curie temperatures clearly indicates titanomagnetite as the predominant magnetic carrier. This conclusion is supported by both the optical observations and thermal unblocking temperatures.

Table 3. Curie point determinations.

Pleasant Mountain, Quartz Syenite Member	563°C
Burnt Meadow Complex, Quartz Syenite Member	565°C
Cuttingsville Intrusion, Essexite	570°C
Randall Mountain, Volcanic Member	572°C
Little Rattle Snake Intrusion, Volcanic Member	356°C, 580°C
Gassetts Lamprophyre	556°C

Chemical remanent magnetization

Any sample could have acquired a chemical remanent magnetization (CRM) due to changes in the magnetic minerals by chemical means after it had passed below the thermal blocking temperature. Evidence of hydrothermal alteration is present in a part of the Little Rattlesnake Complex containing abundant pyrite. Petrographic observations showed that there was a local zone of hydrothermal alteration that resulted in the magnetite component of the titanomagnetites being replaced by pyrite, whereas the ilmenite lamellae were not altered (Fig. 9). This reaction results in the formation of a non-magnetic rock.

A possible secondary magnetization could be produced by later oxidation alteration of the titanomagnetite grains in a fluid-rich environment at elevated temperature. It would appear that such an oxidation would require significantly elevated temperatures, in excess of the temperatures required for oxygen isotopic diffusional exchange within magnetite crystals. Experiments on oxygen isotopic diffusion show that it is very slow below 300 °C (Valley & Graham 1991). Petrologic evidence does not support the elevated temperatures, in excess of 300 °C, necessary for such diffusion after the emplacement and cooling of the intrusions. Furthermore, fission tracks in apatite commonly anneal at about 110 °C and fission-track ages of apatite in the region indicate that none of the rocks has been above this temperature since the late Cretaceous. There is no evidence for elevated temperatures with associated chemical reactions and isotopic exchange that could have produced a later chemical remagnetization since the original cooling of the intrusions.

Conclusions on magnetic mineralogy and stability

By evaluating the data from thermal and AF demagnetization, rock-magnetic properties and optical observations indicating the absence of post-cooling alteration, it is concluded that the majority of the remanent directions obtained in this study are a primary TRM. The Curie temperatures, which fall in a discrete interval between 556 °C and 580 °C, the thermal unblocking temperatures, between 550 °C and 580 °C, and the optical observations all demonstrate that the predominant magnetic mineral is titanomagnetite. Titanomagnetite occurs as discrete C1–C3 grains, as exsolution lamellae in the silicates and as fine reaction rims on the silicates that were produced by magmatic crystallization, by post-magmatic cooling or by deuteric oxidation alteration during post-magmatic cooling.

Palaeomagnetic results

The site-mean directions for the Burnt Meadow intrusions (10 sites), the Pleasant Mountain intrusions (five sites), and the Little Rattlesnake intrusions (12 sites) in the 113–111 Ma group are listed in Table 1 and shown in Fig. 10. The palaeomagnetic results from the Little Rattlesnake intrusions indicate that both syenite and porphyry were in their present relative positions before cooling through their Curie temperatures. Except for one outlier from Burnt Meadow, these intrusions show good directional clustering of the data, no evidence of tilting within the intrusions after cooling and no evidence of tectonic tilting between the intrusions, even though they are separated by distances as great as 130 km.

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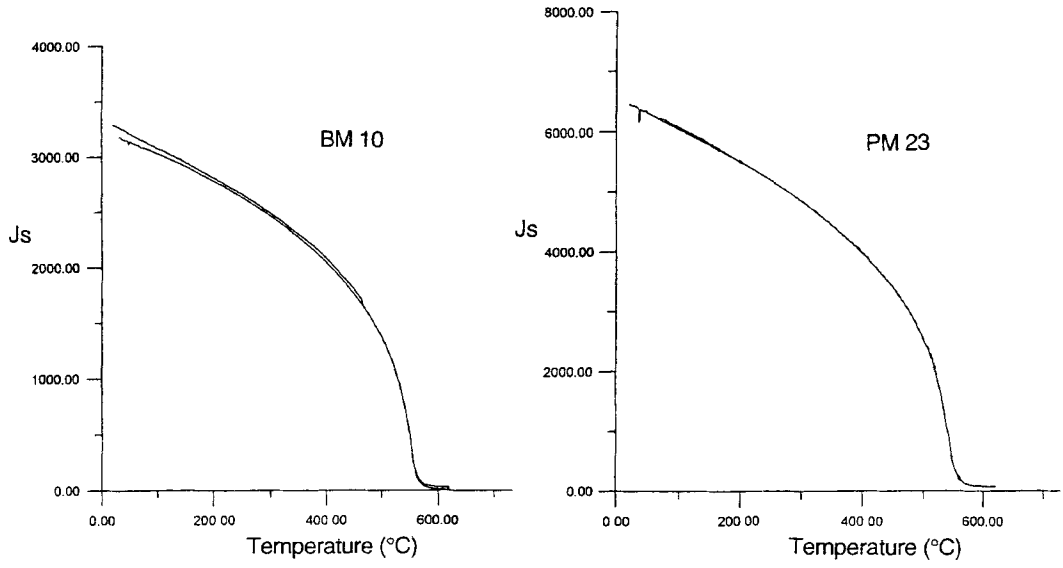


Figure 8. Diagram of reversible Curie temperature runs on C2 titanomagnetites from Pleasant Mountain and Burnt Meadow Complex. No oxidation or alteration to the titanomagnetite was observed during the run. Applied field versus Temperature (°C).

The site-mean directions for the Cuttingsville intrusions (six sites), the Randall Mountain intrusion (11 sites), and the Gassetts lamprophyre (one site) in the 103–100 Ma group are listed in Table 2 and shown in Fig. 10. The within-site precision is good. The agreement between intrusions is also good. Again, there is no evidence of tilting within the intrusions after cooling and no tectonic tilting between the intrusions, even though they are separated by distances as great as 170 km across the strike of known Triassic–Jurassic faults. 17 of the 18 sites were used in calculating the 100 Ma pole. Site RM 10 was excluded because although it had an a_{95} of only 6.8°, it possessed very high NRM values (>100 A m^{-1}) and displayed erratic demagnetization behaviour. One site from the Cuttingsville intrusion had an $a_{95}>20°$, but exclusion of the data did not affect the mean. This site was retained and used in calculating the intrusion site-mean direction.

In the 113–111 Ma intrusions, two sites were unstable, LRS 8 and LRS 9, and had $a_{95}>20°$. These two sites were not used in calculating the mean direction. Except for one of the 18 sites in the 103–100 Ma group and two sites excluded from the 26 sites in the 113–111 Ma intrusions, all other sites show excellent intrasite agreement (Fig. 11). These data are combined with 41 sites (237 samples) from the dual polarity 120–124 Ma southern Maine intrusions (McEnroe 1996b). The 124–120 Ma intrusions show excellent agreement within each polarity and pass a reversal test. When all 85 sites are plotted (Fig. 11) there is little scatter between the means of the sites, although these data represent approximately 25 Myr and are from intrusions spread across 180 km of New England.

Comments on palaeosecular variation during the Early to Middle Cretaceous

Because these complexes were intruded over a period of approximately 25 Myr, the angular dispersion (S) of the site-mean directions or the site-mean VGPs may provide an estimate of the magnitude of palaeosecular variation (PSV) during the Lower Cretaceous. Except for the Gassetts dyke and the Lebanon diorite, the complexes sampled represent several discrete igneous cooling units. The sampling strategy was designed to collect samples from as many cooling units as possible. When the complete data set of an intrusive complex is viewed, it is expected that a mean direction will be obtained that is free from the effects of temporal variations. However, the angular dispersion measurements from individual cooling units, which may record PSV, are markedly lower than expected, and these results are presented here from that viewpoint.

The estimate of the angular dispersion is given by $S=81°/K^{-1/2}$ (Watson & Irving 1957), where K is equal to the precision parameter of the site-mean VPGs. For the 124–120 Ma group of intrusions the angular dispersion of 41 site-mean VGPs is $S=8.8°$, with upper and lower limits at the 95 per cent confidence level of $S_u=10.4°$ and $S_l=7.6°$ using the method of Cox (1969). For the 27 VGPs from the ~113–111 Ma intrusions, the angular dispersion is 10.9° with upper and lower limits of $S_u=13.4°$ and $S_l=9.2°$. The 17 VGPs from the ~103–100 Ma intrusions yield a higher $S=11.9°$ with upper and lower limits of $S_u=15.6°$ and $S_l=9.6°$. Overall, the angular dispersion is still markedly smaller than the predicted value (~15°) for a calculated palaeolatitude of ~40° (Merrill & McElhinny 1983). Butler (1992) has suggested that the Cretaceous normal-polarity superchron was a period of low palaeosecular variation (approximately 75 per cent of present-day values). Thus the estimates of PSV from the 113–111 Ma and 103–100 Ma normal-polarity intrusions support Butler's proposition, indicating that palaeosecular variation was approximately 75 per cent of the present-day value. It is also interesting to note that the dual-polarity Early Cretaceous 124–120 Ma intrusions record an S value that is approximately 60 per cent of the present-day value. The angular dispersions calculated here may indicate that there was lower PSV in the Lower Cretaceous, a phenomenon not restricted to the normal-polarity superchron.

Palaeopoles and comparisons with other Cretaceous data

High-precision palaeomagnetic reference poles are needed more as methods and modelling techniques improve.

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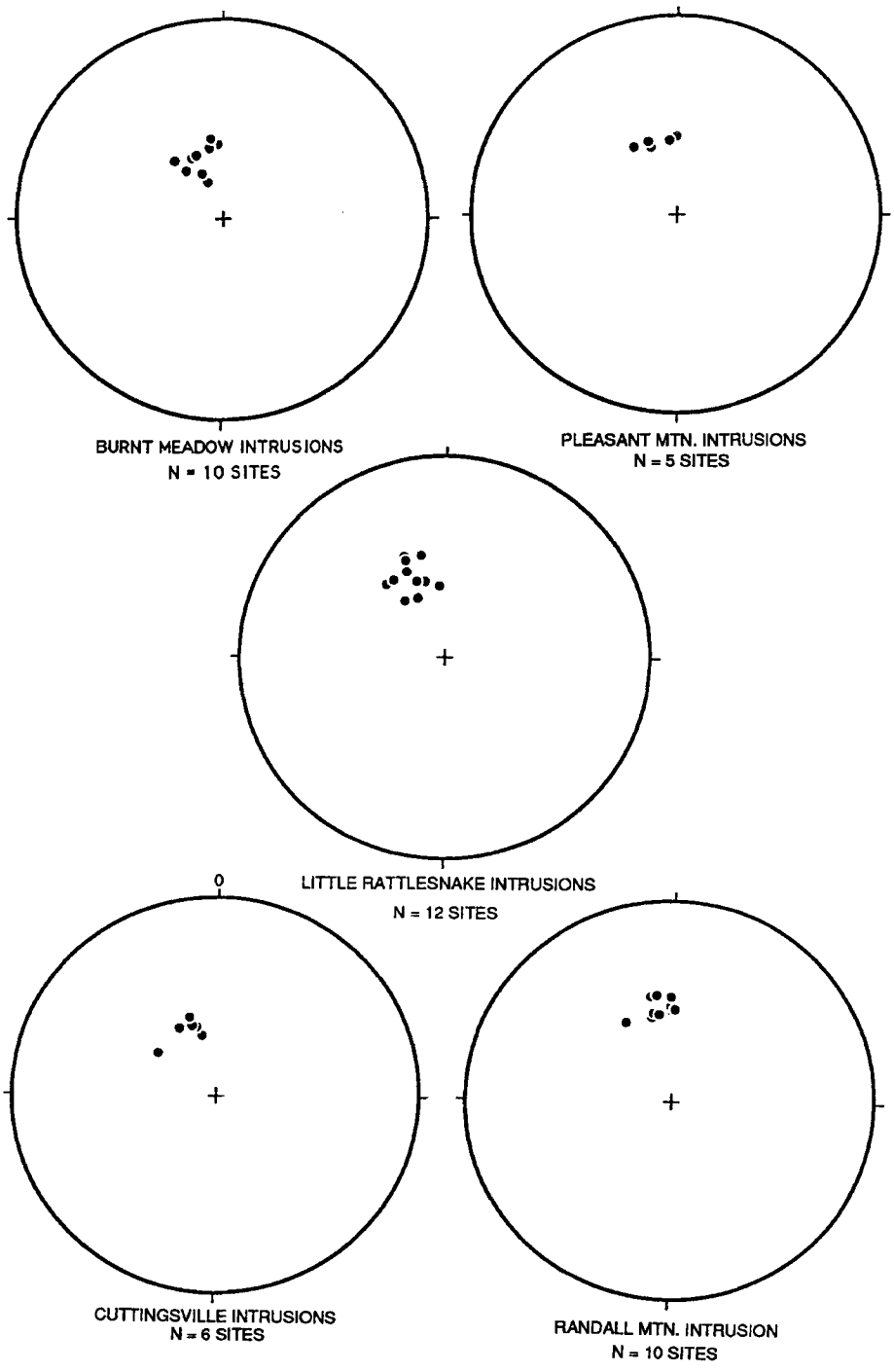


Figure 10. Equal-area diagram of site-mean palaeomagnetic directions from the Burnt Meadow, Pleasant Mountain, Little Rattlesnake, Cuttingsville and Randall Mountain intrusions. Closed circles: lower-hemisphere projections; N: number of sites.

Differences of only a few degrees can be crucial to plate-motion studies. Older techniques (Irving & Irving 1982) using the sliding-window method averaged out sharp changes in plate motion. The cusps present in Mesozoic APW paths (May & Butler 1986) are the result of new modelling techniques and increased precision in palaeomagnetic poles. Synthetic models (Gordon *et al.* 1984) constrain the path to small circles. Well-dated and high-precision data are needed to determine whether plates are constrained to small-circle rotations and, if so, the length of each small-circle track.

The calculated palaeopoles (Fig. 12) are 74.5 °N, 195.2 °E, $A_{95} = 3.8°$ for 113–111 Ma and 76.6 °N, 167.5 °E, $A_{95} = 5.3°$ for 103–100 Ma. The mean palaeopole for the 113–100 Ma normal-polarity group is 76.7 °N, 185.6 °E, $A_{95} = 3.2$, and, when combined with the 124–120 Ma group palaeopole at 72.3 °N, 198.2 °E, $A_{95} = 3.3°$, the resultant palaeopole for all 85 sites is 74.2 °N, 194.0 °E, $A_{95} = 2.3°$. These palaeopoles fall north of the Jurassic–Cretaceous track of the Mesozoic APW path by May & Butler (1986) based on only eight studies. The reliability criteria for palaeopoles that they employed are as follows.

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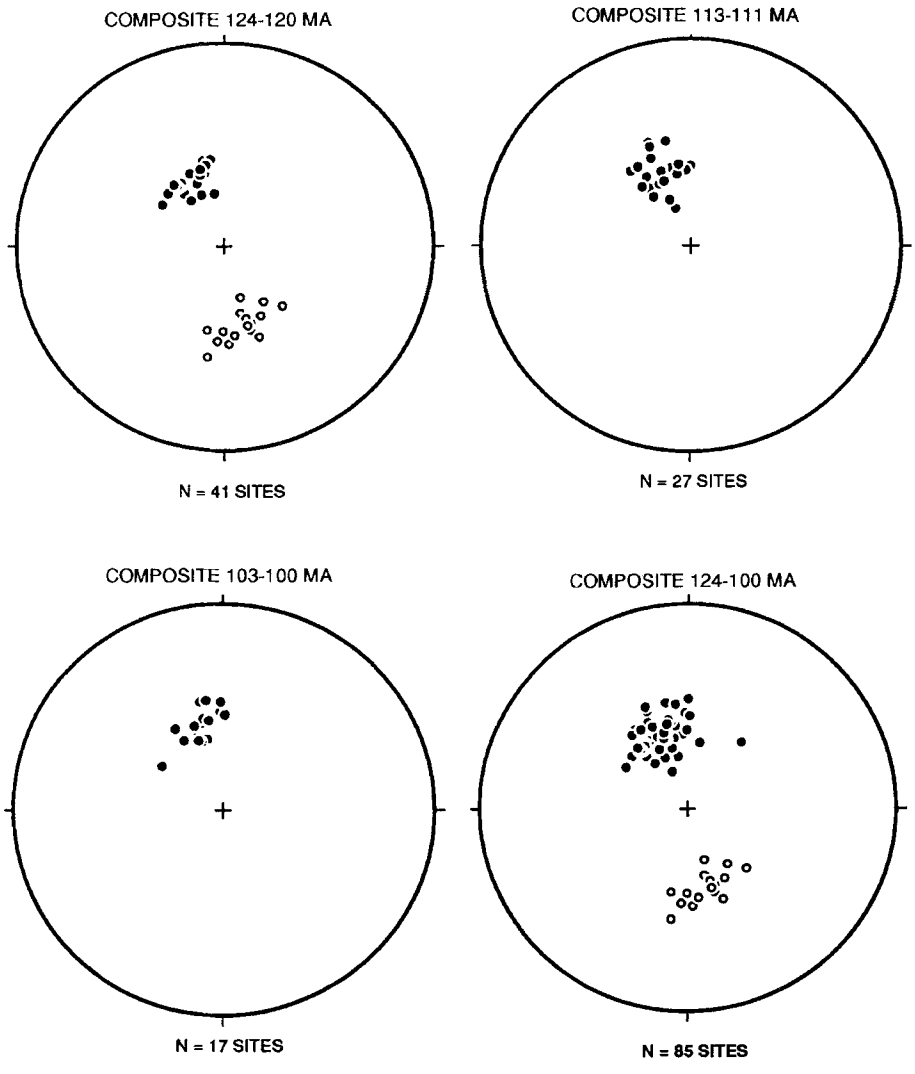


Figure 11. Equal-area diagrams of site-mean palaeomagnetic directions from the 124–120, 113–111 and 103–100 intrusion groups, and a composite diagram showing the palaeomagnetic directions from all 85 sites during the 124–100 Ma interval. Open circles indicate upper-hemisphere projections; closed circles, lower-hemisphere projections. N: number of sites.

(1) A primary magnetization was isolated.
(2) Age was known ± 10 Myr.
(3) The number of sites was greater than or equal to 10.
(4) The angular-dispersion precision parameter k was greater than 20 and less than 150, because high k values commonly fail to average out secular variation. The one exception to this is slowly cooled igneous bodies. Conversely, values of k less than 20 suggest the presence of an unremoved secondary component.
(5) An a_{95} that was less than or equal to 15°.

Van der Voo (1993) has expanded the criteria for reliable palaeopoles. He has developed a quality index (Q) which includes a well-determined age of the rocks, stability tests, tectonic coherence, the presence of reversals, and no suspicion of remagnetization. This combined study exceeds the criteria of May & Butler, and meets all the criteria of Van der Voo (1993), giving the maximum Q of 7.

Listed in Table 4 are other North American Cretaceous palaeopoles. It is important to note that the Early Cretaceous Monteregian Hills (Foland *et al.* 1986) pole (Foster & Symons 1979) was not used in previous models, although it has a very high Q of 6, whereas the Isachsen diabase (Larochelle, Black & Wanless 1965) and Mount Ascutney (Opdyke & Wensink 1966) poles were used, and have low Q values of 3. Although both the Niobrara Formation (Shive & Frerichs 1974) and the Notre Dame Bay dykes (Lapointe 1979) have high Q values, they also have high values of k (170 and 180, respectively), which could indicate the presence of an unremoved secondary overprint, and, if used, should be used with caution.

The coherence in pole positions from New England (this study), the Monteregian Hills, and the Arkansas intrusions is striking. At a first glance it would appear that there has been no APW relative to North America for an interval of about 30 Myr, from 120 to 86 Ma. If one looks more closely at the data, they could suggest continual, though small, APW relative to North America in this time span. The earliest Cretaceous palaeopoles, southern Maine, the Monteregian Hills and the Notre Dame Bay dykes, have lower-latitude palaeopoles than

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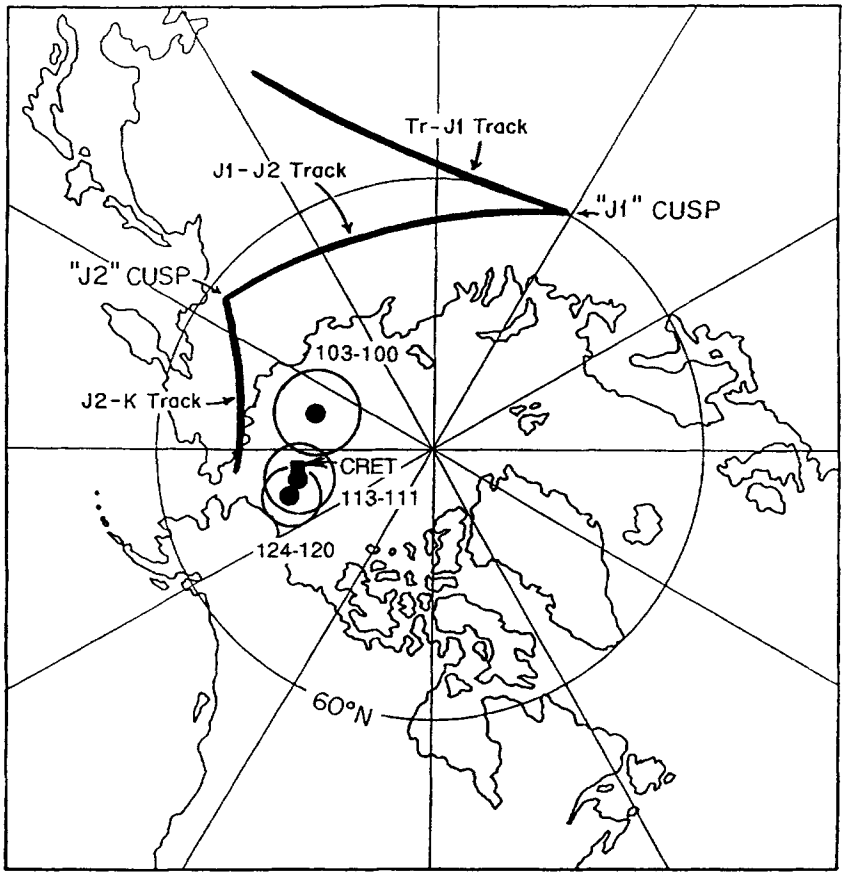


Figure 12. Mesozoic North American apparent polar-wander path of May & Butler (1986) showing palaeomagnetic poles (small black circles) with A_{95} for the 103–100 Ma, 111–113 Ma and 124–120 Ma intrusions. The mean palaeopole for the 124–100 Ma Cretaceous group with A_{95} is shown with a black square.

the 113–100 Ma New England palaeopoles (this study) and those from Granite Mountain, Arkansas (Globerman & Irving 1988). The Late Cretaceous palaeopoles, listed in Table 4, from Elkhorn Mountains (Diehl 1991) and Adel Mountains, Montana (Gunderson & Sheriff 1991) have, on average, a significantly higher latitude than previously determined Early to Middle Cretaceous palaeopoles. Diehl (1991) has calculated a 9.4° arc of separation between the mean Cretaceous pole of Globerman & Irving (1988) and the Elkhorn Mountains pole. Gunderson & Sheriff (1991) and Diehl (1991) have suggested that the onset of APW relative to North America occurred during the late Cretaceous, prior to 80 Ma, and was rapid. Several of these authors have associated this rapid onset of APW with the start of the Laramide orogeny and an increase in absolute motion of the North American plate in the hot-spot reference frame (Engebretson, Cox & Gordon 1985). When these new 113–100 Ma poles from New England are combined with the Arkansas poles (101–86 Ma), this yields a palaeopole of 185.1 °E, 75.7 °N, $A_{95} = 4.5$ °, and hence only a 5.0° arc separation from the Elkhorn Mountains. At the 95 per cent confidence level, this new combined palaeopole is indistinguishable from the late Cretaceous Adel Mountains and the Elkhorn Mountains poles. This would suggest that the Laramide orogeny did not have a rapid start-up, but that it was a more gradual process.

As has been suggested, there appears to have been a slow-down in North American plate motion in the Early Cretaceous, and a virtual standstill until the Middle to Late Cretaceous, when plate motion increased. This study supports a gradual increase in plate motion to the north-west from Middle to Late Cretaceous. When these Cretaceous palaeopoles are combined with the mid-Jurassic poles from the White Mountains (Van Fossen & Kent 1990) and from central Massachusetts (McEnroe 1996a) and Connecticut (McEnroe 1991), the validity of the J2 cusp of May & Butler comes into question. The mid-Jurassic high-latitude palaeopoles determined by McEnroe (1991) and Van Fossen & Kent (1990) are consistent with these new Cretaceous poles, suggesting a higher-latitude and smoother APW path for North America from the mid-Jurassic to the mid-Cretaceous.

The Cretaceous average determined by Mankinen (1978), which was used by May & Butler (1986), and the synthetic pole calculated by Gordon *et al.* 1984), which used older data with poorer Q factors than recently determined poles, should not be used without elimination of the poorly determined poles from the averages and the addition of recently determined high-precision poles. The importance of a reliable Cretaceous reference pole for North America is crucial to reconstructions of the paths and rotations of displaced terranes of the Western Cordillera of North America. The study of microplate tectonics in the Cordillera has been seriously hampered by the limited North American palaeomagnetic data for the mid-Jurassic and

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Table 4. Cretaceous North American Palaeopoles.

	Age	Plat°N	Plong°E	K	a_{95}/A_{95}	S	Q	Reference
Notre Dame Bay Dikes	128	67	212	180.0	4.0	6.0	5	1
Monteregian Hills	122 ± 2	72.4	169	49.2	2.4	11.6	6	2
Normal polarity		73.4	163.9	37.6	3.6	13.2	5	2
Reverse polarity		70.7	175	99.7	2.7	8.11	5	2
White Mountain Average	112-121±5	71.9	187.4	322	6.9	4.5	5	3
Mount Ascutney	119 ±4	68.5	186.6	190	14.0	5.9	4	3
Pawtuckaway Mtn.*	121 ±4	76.9	191.2	N/A	N/A	N/A	3	3
Tripyramid Mtn.	112 ±5	70.2	185.7	282	7.4	4.8	3	3
Southern Maine								
Lebanon Diorite	125 ±3.0	73.2	195.1	104	7.5	7.6	5	4
Tatnic Complex	122 ± 2.0	65.7	208.4	113	4.6	7.6	5	4
Alfred Complex	120.4 ±1.0	74.0	209.8	122	3.6	7.4	5	4
Cape Neddick	120.6 ±0.4	76.2	175.2	91	4.6	8.5	5	4
Burnt Meadow Complex	112.6 ±0.2	75.7	208.5	98	5.6		5	This study
Pleasant Mtn.	112.1 ±0.4	77.4	185.0	141	6.5	6.8	5	This study
Little Rattlesnake Complex	111 ±3.0	71.5	182.6	96	4.4	8.3	5	This study
Randall Mtn.	103 ±4.0	76.5	140.0	145	4.0	6.7	5	This study
Gassetts Lamprophyre*	100 ±2.0	74.0*	178.9*	32	10.9	-	-	This study
Cuttingsville Complex	100 ±0.3	72.1	197.0	87	8.3	8.7	5	This study
Combined New England								
Southern Maine	124 - 120	72.3	198.2	85	3.3	8.8	6	4
New England	113 - 111	74.5	195.2	56	3.8	10.9	6	This study
New England	113 - 100	76.7	185.6	49	3.2	11.6	6	This study
New England	103 - 100	76.6	167.5	47	5.3	11.9	5	This study
New England	124 - 100	74.2	194.0	46	2.3	11.9	7	This study
Isachsen Diabase	109 ± 5.0	68	183	20	8.0	18.1	3	5
Niobrara	84 - 74	65	198	170	4.0	6.2	5	6
Arkansas Combined	101 - 86	74.1	192.5	34.0	5.7	13.9	5	7
Magnet Cove	101 - 97	72.6	197.3	42.3	6.0	12.5	5	7
Granite Mtn.	91 - 86	77.8	171.4	21.0	17.1	17.7	5	7
Elkhorn Mtn	80.8 ± 2.4	80.3	189.5	16.8	9.6	19.2	5	8
Adel Mtn.	76	82.2	209.9	18.4	6.8	18.9	4	9
Roskruge Volcanics	72	70	177	86	6.0	8.7	5	10
Cretaceous Mean	130 - 85	68	186	N/A	2.2	N/A	-	11
Cretaceous Synthetic	126 - 85	69.9	170	N/A	N/A	N/A	-	12
Middle Cretaceous	-	71	196	N/A	4.9	N/A	-	13
Paleocene Intr, MT	69 - 60	82	181	20	5.0	18	6	15

*VGP Plat, Plong: palaeolatitude (°N) and palaeolongitude (°E) of magnetic pole; a_{95*}: radius of 95 per cent cone of confidence or A_{95}; k: statistical parameters of Fisher (1953); S: angular dispersion; Q: quality index of Van der Voo (1993). N/A: not available or cannot be calculated from published data. References: (1) Lapointe (1979), (2) Foster & Symons *et al.* (1979), (3) Van Fossen & Kent (1992), (4) McEnroe (1996), (5) Larochelle *et al.* (1965), (6) Shive & Frerichs (1974), (7) Globerman & Irving (1988), (8) Diehl (1991), (9) Gunderson & Sheriff (1991), (10) Vugeteveen, Barnes & Butler (1981), (11) Mankinen (1979), (12) Gordon *et al.* (1984), (13) Globerman & Irving (1988), (14) Diehl *et al.* (1983).

Cretaceous. The data set presented here should be of aid in such studies.

CONCLUSIONS

Well-constrained Aptian to Albian palaeopoles are reported here from precisely dated 113–100 Ma intrusions in New England. The remanent magnetizations are of normal polarity. High-unblocking-temperature magnetizations isolated from 44 of the 47 sites are shown to be primary. The primary nature of the remanence carrier has been verified by detailed palaeomagnetic and rock-magnetic experiments, and by optical observations. The dominant remanence carrier is titanomagnetite, which crystallized from the magma or was produced by deuteric oxidation alteration during post-magmatic cooling. The calculated palaeopoles are 74.5 °N, 195.2 °E, $A_{95} = 3.8°$ (113–111 Ma) and 76.6 °N, 167.5 °E, $A_{95} = 5.3°$ (103–100 Ma). The mean palaeopole for the 113–100 Ma normal-polarity group is 76.7 °N, 185.6 °E, $A_{95} = 3.2°$, and when combined with the Barremain and Albian 124–120 Ma dual-polarity southern Maine intrusions, with a palaeopole at 72.3 °N, 198.2 °E, $A_{95} = 3.3°$, the resultant 124–100 Ma palaeopole is 74.2 °N, 194.0 °E, $A_{95} = 2.3°$. These newly determined palaeopoles all plot north of, and at higher latitude than, the predicted Jurassic–Cretaceous track for North America (Gordon *et al.* 1984; May & Butler 1986) indicating that a revision of this portion of the Mesozoic APW for North America is needed. The estimated angular dispersions calculated support lower palaeosecular variation values in the Early Cretaceous, and may not be limited only to the Cretaceous normal-polarity superchron.

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